IEEE 802.11ba

Wake-Up Radio

Solving IoT's Power Paradox

THE PROBLEM

A $120 Smart Smoke Detector

Features

Connects to WiFi, detects smoke, sends notifications to your phone—everything you'd expect from a modern IoT device.

The Fatal Flaw

When the battery dies, you cannot replace OR recharge it.

The real power drainer? The Radio Device (WIFI).

THE CHALLENGE

The Engineering Tradeoff

IEEE 802.15.4 Standard

A low-power wireless standard designed for IoT devices.

Data Rate

250 kbps

Maximum speed

Range

~ 10 meters

Not enough for a house

The Fundamental Tradeoff

"The more transmission power you use, the longer your range. But more power drains your battery faster. You cannot have both high energy efficiency AND high data rates **with a single radio** ."

Energy Efficiency VS High Performance

THE BREAKTHROUGH

Let There Be No Signle Radios

If one radio can't do it...

Use TWO radios!

CHAPTER 01

WUR Radio

High-Level Architecture

PRIMARY RADIO

PCR: Primary Connectivity Radio

PCR

High Speed Data Transmission

Capable of handling large data payloads at high speeds

High Power Consumption

Requires significant energy to maintain connectivity

High Range

Covers large distances throughout your entire home

Must sleep to save power—but who wakes it up?

COMPANION RADIO

CCR/WUR: The Wake-Up Radio

Extremely Low Power

Consumes less than **1 mW**—negligible energy footprint

Always Awake

Never sleeps—constantly listening for incoming signals

Receiver Only

Cannot transmit—dedicated purely to listening

One Job: Wake Up PCR

Detects wake-up frames and physically signals the PCR to power on

WORKFLOW

How They Work Together

Scene: 1 AP (WiFi Router) + 1 STA (Your Device with 2 Radios)

1. AP has data to send — PCR asleep
2. AP sends wake-up frame — To WUR
3. WUR detects frame — Always on
4. WUR signals PCR: "Wake up!" — Physical signal
5. PCR powers on — "I'm ready"
6. AP sends data to PCR — High speed
7. PCR goes back to sleep — Save power

The PCR only consumes power when actively transmitting data—otherwise, it's asleep!

CHAPTER 02

IEEE 802.11ba Standard

Technical Implementation

STANDARDIZATION

IEEE 802.11ba Evolution

2017

First Draft Released

Initial specification for Wake-Up Radio technology introduced to IEEE working group

2018-2020

Draft Iterations

Multiple revisions and refinements based on industry feedback and testing

2021

Official Release

IEEE 802.11ba officially published and ready for industry adoption

POWER OPTIMIZATION

Duty Cycling: Let WUR Sleep Too

The WUR's "Work Day"

1 Minimum Wake-Up Time

Minimum duration WUR must stay awake during a duty cycle

2 Duty Cycle Period

How long is one complete work day (awake + sleep)

3 Length of "Awake" Period

How long does WUR stay awake during a work day

4 Starting Time of "Awake" Period

When does the awake period begin (synchronization)

The Negotiation Process

1. AP advertises Min Wake-Up Time
2. STA replies with Duty Cycle & Awake Length
3. AP replies with Starting Time
4. ✓ Negotiation Complete!

SYNCHRONIZATION

WUR Beacon: Maintaining Connection

Normal 802.11

Beacon Frames

AP broadcasts periodically with 64-bit timestamp

- **Synchronize** all devices to common clock
- **Heartbeat:** "You're still connected"

Problem: PCR loses sync when asleep!

WUR Solution

WUR Beacon Frame

New frame type introduced specifically for wake-up radios

- AP periodically transmits **WUR beacons**
- Provides "**heartbeat**" function
- **Compressed** timestamp (not 64-bit)

Compressed format saves transmission time—WUR needs to sleep!

NETWORK MAINTENANCE

Re-Discovery: Finding Your AP

Normal 802.11

If device stops receiving beacon frames, it assumes it's out of range and needs to find a new AP.

Re-Discovery Process:

1. Scan ALL available channels
2. Find one or more APs
3. Authenticate with AP
4. Associate with BSS

Extremely power hungry!

NETWORK MAINTENANCE

Re-Discovery: Finding Your AP

WUR Solution

Engineer's optimization: Dedicate a single channel for discovery instead of scanning all channels.

- All APs send **WUR Discovery Frames**
- Contains: **AP address + operating channel**
- Device monitors **ONE channel** only
- It picks one AP to **authenticate and associate** to

Wake-Up Radio technology **breaks the fundamental power-range tradeoff**, enabling always-connected devices that last for years on a single battery.

IEEE 802.11ba represents a paradigm shift in wireless communication design

Ultra Low Power
<1 mW consumption

Full Range
Whole-home coverage

Years of Life
Extended battery life

References

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